

Journeyman Final Exam Answers

Journeyman Final Exam #4 - ANSWERS

- | | |
|--------------------------------------|---|
| 1. (a) largest two | 220.56 |
| 2. (d) 135% | 460.8(A) |
| 3. (c) Table 1 | 356.22 |
| 4. (d) four | 220.14(I) |
| 5. (a) 10 cubic inches | Table 314.16(A) |
| 6. (d) 0.10Ω | Table 9 |
| 7. (c) nominal | DEF 100 |
| 8. (d) 40 amps | 440.62(A)(2) |
| 9. (c) 200 amps | Table 392.60(B) |
| 10. (a) one is required | 210.52(D) |
| 11. (b) 3"x 2"x 2 1/4" | Table 314.16(A) |
| 12. (d) 200va | 220.14(G)(2) |
| 13. (b) 13 receptacles | 605.9(C) |
| 14. (b) 20 amps | 430.53(A) |
| 15. (d) one and two family dwellings | 210.52(E)(1) |
| 16. (b) clock in a dining room | 210.52(B)(2) ex.1 |
| 17. (b) 800 amperes | 240.4(B)(3) |
| 18. (d) 30 volts | 522.28 |
| 19. (c) I & II | 220.53 |
| 20. (a) lowest | 310.14(A)(2) |
| 21. (d) 2 1/8" | Table 314.16(B)(1) 5 x 2.25 cu.in. = 11.25 and
4 x 2 cu.in. = 8 11.25 + 8 = 19.25 cu.in.
Table 314.16(A) = 2 1/8" |
| 22. (b) | Table 4 - 1" EMT = .346 area |
| 23. (a) 11.52 kW | Table 220.55 (Col. B) 32 kW x 36% = 11.52 |
| 24. (d) 16 amps | Table 310.15(C)(1) #14 THW = 20a x 80% = 16 |
| 25. (a) 11 kW | Table 220.55 Col.C = 11 kW for two appliances which
is the smallest value |
| 26. (b) 17.25 amps | Table 430.248 13.8a x 125% (430.22) = 17.25a |
| 27. (a) 6000 VA | 220.5(C) living area 2000 sq.ft. x 3va (220.41) = 6000 |
| 28. (c) 4500 VA | 220.52(A,B) 3000 VA + 1500 VA = 4500 VA |
| 29. (b) #4 | Table 250.66 |
| 30. (c) 22.5 amps | Table 430.248 10a x 125% 430.24 = 12.5a + 10a = 22.5 |
| 31. (c) 21.6 amps | #12 TW = 20a x 1.08 (T.310.15(B)(1)(1) correction factor) = 21.6 |
| 32. (c) 60 amps | Table 430.248 34a x 175% Table 430.52 = 59.5 = 60
(430.52 ex.1) (240.6) |
| 33. (d) 16 amps | #14 THW = 20a x 80% Table 310.15(C)(1) = 16 |
| 34. (c) 22.1 amps | T.430.248 17a x 130% (430.32(C) maximum) = 22.1 |
| 35. (b) .82 | Table 310.15(B)(1)(1) correction factors |

Half way through the exam
(Answers continued next page)

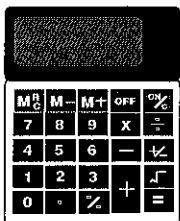
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36. (a) 6 kW Table 220.55 Note 4 Branch Circuit - Nameplate rating for **one**
37. (d) 33.6 amps Table 430.248 24a x 140% (430.32(C) maximum) = 33.6
38. (d) 18 cu.in. 314.16(B) 2-white, 2-black, 1-bare, 1-clamps, 2-duplex = 8 wires
Table 314.16(B) 8 wires x 2.25 cubic inch = 18
39. (c) 35 amps Table 430.250 28a x 125% (430.22) = 35
40. (a) 21 amps Table 310.16 #12 THHN = 30a x 70% Table 310.15(C)(1) = 21
41. (b) 25 amps no derating required in circuit
42. (c) 33.75 kW 10 x 4.5 kW = 45 kW x 75% (220.53) = 33.75 kW
43. (a) 30 amps Table 310.16 footnotes * overcurrent protection
44. (b) full load 220.82(C)(1)
45. (b) 7.15 kW Table 220.55 5 kW + 8 kW = 13 kW x 55% (Col.B feeder) = 7.15
46. (d) 15.7 kW T.220.54 6 x 5 kW = 30 kW x 75% = 22.5 kW x 70% (220.61(B)) = 15.7
47. (d) 5 2500 sq.ft. x 3va = $\frac{7500va}{3} = 4.16$ or 5 circuits
15 a x 120v = 1800va
48. (d) none of these 180va does **NOT** apply to a residence (dwelling)
49. (b) 16.8 kW Table 220.55 15 x 3.5 kW = 52.5 kW x 32% (Col.B) = 16.8 kW
50. (a) 0.0209 Table 5
51. (c) 6.4 kW Table 220.55 Column B 8 kW x 80% = 6.4 kW
52. (d) 5400va Section 220.5(C) outside dimensions 1800 sq.ft. x 3va = 5400va
53. (a) 20 amp
mits T.430.248 13.2a x 175% (T.430.52(C)(1) = 23.1a 430.52 ex.1 per-
going up to the **next higher** standard size. The next higher per Table
240.6 is a 25 amp not 30. •But, 25 amp is not a choice!
54. (b) 6.75 cu.in. 314.16(B)(1,5) 3 x 2.25 cu.in. T.314.16(B)(1) = 6.75
55. (c) #8 Table 250.122
56. (c) 17.16a Table 430.248 13.2a x 130% (430.32(C) maximum) = 17.16
57. (d) .3802 sq.in. Table 5 - #8 XHHW = .0437 x 6 = .2622
#6 XHHW = .0590 x 2 = .118
.3802
58. (c) 79 amps •The maximum on the neutral would be if line one is off $9500w \div 120v = 79a$
59. (c) .1921 sq.in. Table 4 - 1" rigid metal = .355 Table 5 - #12 XHHW = .0181 x 9 = .1629
.355 - .1629 = .1921
60. (d) 13 Table 4 - 1" IMC = .959 (100% csa) x 60% (nipple fill) = .575
Table 5 - #8 TW = .0437 .575 ÷ .0437 = 13

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61. (c) 150%	Table 430.52(C)(1)
62. (c) 19	Table 4 - 2" PVC 80 = 1.150 Table 5 - #6 XHHW = .0590 $1.150 \div .0590 = 19$
63. (b) 15.225 amps	#14 RHH = 25a x 70% T.310.15(C)(1) = 17.5a x .87 T.310.15(B)(1) correction factor = 15.225
64. (c) 14.25 cu.in.	Table 314.16(B)(1) #12 = 2.25 cu.in. x 3 = 6.75 cu.in. #10 = 2.5 cu.in. x 3 = <u>7.5 cu.in.</u> 14.25 cu.in.
65. (a) 2"	Table 5 #3 THWN .0973 x 6 = .5838 #8 THW .0437 x 3 = .1311 #10 THW .0243 x 2 = <u>.0486</u> .7635
66. (a) 12	Table 4 1" nipple 60% fill = <u>.532</u> = 12.1 or 12 conductors Table 5 #8 THW = .0437
67. (d) 45 amps	17a F.L.C. x 250% T.430.52(C)(1) = 42.5a Table 240.6 = 45a
68. (c) 15	Table 4 1 1/2" IMC = .889 ÷ .0590 Table 5 #6 XHHW = 15
69. (a) 43 kW	T.220.55 Column C = 15 kW + 28 kW = 43 kW
70. (d) 17,000va	Table 430.250 46.2a F.L.C. x 208v x 1.732 = 16,644va.



After completing the exam, grade yourself by counting the correct number of questions and divide by the number of questions answered.

Example: 53 correct answers $\div$ 70 questions = 76%.